

RESULTS

The relative abundance of size classes 1-4 illustrates annual dynamics at each location (Figure 4-2). Through time, the most general pattern of change was a disproportionate reduction of larger size classes in location-years exposed to significant storms. In three locations, CUR, FL, and PR, the abundance of the two largest size classes decreased relative to the abundance of the smallest two size classes directly following years with significant storms. In the three locations that did not experience appreciable storm activity, abundance of the larger, reproductive size classes (SC3 and SC4) increased relative to that of the smaller size classes (SC1 and SC2) over the course of the study.

The same qualitative results can be seen when size data are plotted as a continuous rather than a categorical variable (Figure 4-3). The majority of colonies experienced little to no change in size, as evidenced by the preponderance of points lying along the 1:1 diagonal line, with some evidence of increased shrinkage during storm years. Importantly, patterns of reduction in colony size were size-specific, with smaller colonies more likely to lose a larger proportion of their entire size compared with larger sizes (as evidenced by higher variance in the log-log plot at smaller starting size classes in Figure 4-3). Note that only size change of survivors is reported here, with details of whole-colony mortality described along with the matrices below.

Matrices

Direct spatial comparison of transition and fragmentation rates was conducted in one approximately three-year window from 2007 to 2010 across five locations:

three primary study locations, Curaçao, Florida, and Jamaica, where surveys were replicated over several years and patterns can be assessed with greater confidence.

Common response to a temporal event – storms

We witnessed a common effect of storms on adult mortality, population size structure and growth rate. In Chapter 2 we showed an example of storm dynamics in Florida. Here we parameterized storm dynamics in two additional locations, Puerto Rico and Curaçao. In all locations, during storm years the largest adults (SC4) have a higher probability of whole-colony mortality as well as an increased probability of shrinking. Also, relative abundance of large individuals decreases with respect to that of small individuals. This is because large colonies fragment and become smaller, and those fragments stay on the reef. Over this time scale we can also witness the converse: In areas without storms, the relative abundance shifts from small size classes to larger size classes as colonies grow. Where there were enough data after a storm, we witnessed potential recovery (as in the growth from smaller to larger size classes in Curaçao) or lack thereof (as in Florida) (Figure 4-2). Population growth rate was below 1 during storm intervals regardless of location. It may seem obvious that population growth rate decreases during a storm, but considering fragmentation, this was not a foregone conclusion (see Discussion of Lirman 2003 below).

Though reduction in population abundance due to storms was measured, the less-than-decadal time scale of this study is insufficient to capture recovery of a population from a major hurricane. In Florida, two Category 3 or greater hurricanes pummeled the survey area in 2005. The frequency of such an event in a 250 nm radius

palmata), the techniques to achieve these goals and the expected outcomes will be different. In Florida, storms recur frequently, and population growth rates (λ_s) are > 1 in calm years, meaning fragmentation and attachment are occurring on an ongoing basis. Meanwhile, the relative abundance of large individuals has not increased in the six years since the most recent severe storm, meaning that fragmentation of the largest individuals is greater than or equal to growth from smaller size classes. This is likely due to excess nutrients in Florida (compared with Jamaica, for example) (La Pointe 1997). The lack of growth to larger size classes could signify that Florida's population is below some resilience threshold.

In contrast, relative abundance of the larger size classes increased in the two years after the storm in Curaçao (Figure 4-2). Thus, the slow and steady growth with minimal fragmentation that we originally expected to witness in this population could be occurring. Predicted percent cover in this case would be overly pessimistic, as it assumes that a severe hurricane (as experienced in 2008) would occur every six years. We know from hurricane tracks that this frequency is far lower. (Future simulations will account for reduced frequency, and may incorporate matrices typifying other extreme events such as bleaching or disease outbreaks.) Despite rates of population depletion rather than expansion and a significant decrease in percent cover over the survey interval, Curaçao still has the potential to exhibit natural resilience.

Resilience is a system's capacity to absorb disturbance while retaining the self-organizing capabilities to maintain its essential structure, function, and feedbacks (Walker 2009, Hughes et al. 2010). We have highlighted a potential distinction

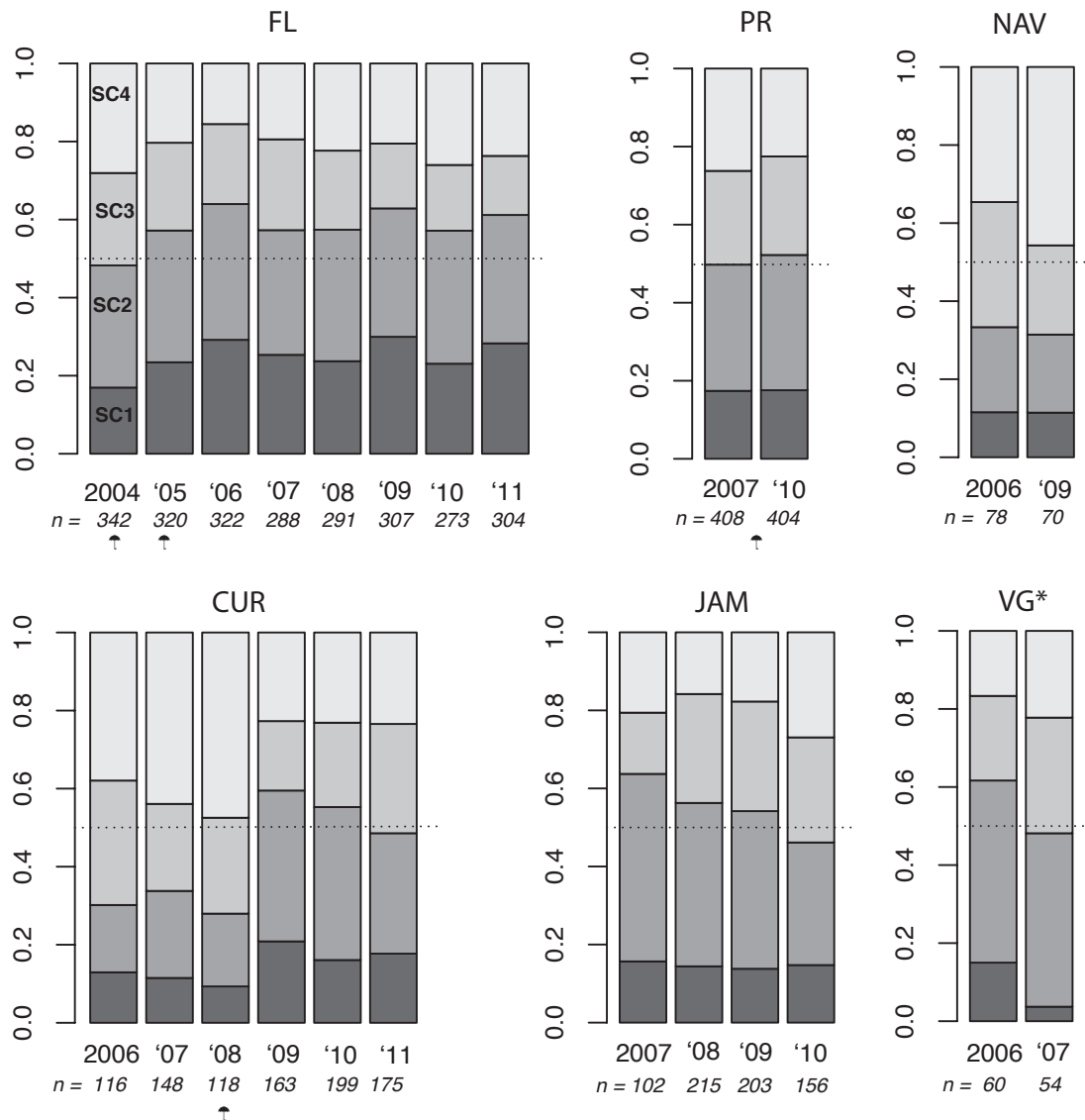


Figure 4-2: Relative abundance per size class per survey for each location. Size classes (SC) 1-4 have average planar dimension of 7, 20, 45, and 100 cm, respectively. In Florida (FL) abundance is based on 15 plots. Nine plots were surveyed in Puerto Rico (PR). Four plots were surveyed in Navasa (NAV). In Curacao (CUR), abundance is calculated from seven plots in 2006 and 2009, and eight plots all other years. In Jamaica (JAM), five plots were surveyed in 2007 and nine plots in 2008-2010. Six plots were surveyed in Virgin Gorda (VG). *In VG new fragments were not tallied, and thus SC1 is underrepresented in 2007. ↑ Umbrella represents years during which significant storm damage occurred.